COVID-19 and the Formality Recovery Path in Peru: A Worker-Level Panel DiD Using Sectoral Teleworkability

Research Pipeline Output
ENAHO Panel Analysis

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Abstract

We estimate the causal effect of COVID-19 on labor informality in Peru using a difference-in-differences design that exploits differential occupational exposure via the Dingel and Neiman (2020) teleworkability index. Using five waves of the ENAHO employment panel (2020–2024, 262,733 observations), we find that contact-intensive workers experienced a 9 percentage point increase in informality relative to teleworkable workers ($p < 0.001$). This effect materialized in 2021 and showed zero attenuation through 2024 (Wald test $p = 0.994$), providing strong evidence of permanent scarring. The teleworkability index is pre-determined and panel attrition is not differential by treatment ($p = 0.163$), strengthening causal identification. The scarring effect is concentrated in the loss of employer-provided social security rather than firm-size changes, suggesting that portable social protection mechanisms may be more effective than traditional formalization strategies.

Keywords: COVID-19, informality, scarring, difference-in-differences, teleworkability, Peru

JEL Codes: J46, O17, J21, I18

Contents

1 Introduction

Peru's labor market informality rate exceeds 70%, one of the highest in Latin America. The COVID-19 pandemic, which triggered strict lockdowns and mobility restrictions beginning in March 2020, disrupted formal employment relationships across all sectors—but its impact was not uniform. Workers in contact-intensive occupations, where physical presence is essential and remote work is infeasible, faced the most severe labor disruptions. This paper asks whether these differentially exposed workers experienced a permanent increase in informality—a "scarring" effect—or whether they recovered to pre-pandemic formality levels by 2024.

We exploit differential sectoral exposure to COVID-19 mobility restrictions using the Dingel and Neiman (2020) teleworkability index as a source of identifying variation. The index, originally constructed from O*NET task descriptions for U.S. occupations, measures the feasibility of performing each occupation remotely. We map this index to Peruvian occupation codes (ISCO-08) via a crosswalk, creating a continuous, pre-determined treatment intensity variable: workers in low-teleworkability occupations (contact-intensive, score < 0.20) constitute our "treated" group, while teleworkable workers serve as the comparison.

Using the ENAHO rotating panel from 2020 to 2024 (Module 500, Employment), we estimate a difference-in-differences event study with individual and year fixed effects. Our main specification interacts year dummies with the teleworkability indicator, yielding differential informality coefficients for each post-shock year relative to the 2020 baseline.

Three main findings emerge. First, contact-intensive workers experienced an immediate and large increase in informality of approximately 9 percentage points relative to teleworkable workers in 2021 ($p < 0.001$). Second, this differential persisted essentially unchanged through 2024, with a Wald test for time-constancy yielding $p = 0.994$—strong evidence of permanent scarring. Third, attrition from the ENAHO rotating panel is not differential by treatment status ($p = 0.163$), strengthening the causal interpretation.

These findings contribute to the literature on labor market scarring (Arulampalam, 2001) and informality dynamics in developing countries (Bosch et al., 2025). While previous work has documented the immediate employment effects of COVID-19 in Peru, our panel-based DiD design provides the first causal evidence on whether the pandemic's formalization shock was transitory or permanent.

The remainder of the paper proceeds as follows. Section 2 reviews the literature. Section 3 describes the data. Section 4 presents the identification strategy. Section 5 reports results. Section 6 discusses robustness. Section 7 concludes.

2 Literature Review

This paper connects three strands of the literature: labor market scarring, informality in developing countries, and the differential impacts of COVID-19 by occupation.

2.1 Labor Market Scarring

The scarring hypothesis holds that adverse labor market shocks can produce persistent effects on workers' employment quality, wages, and formality status long after the initial shock dissipates. Arulampalam (2001) document that unemployment spells in the UK carry a wage penalty of 6% that persists for over a decade. In developing countries, Cruces et al. (2012) show that exposure to economic crises in Argentina durably reduced formal sector attachment. The mechanism is that workers displaced into informality face barriers to re-entry—skill depreciation, loss of employer networks, and stigma—creating a "trap" that outlasts the shock.

2.2 Informality in Peru

Peru has one of the highest informality rates in Latin America, with over 70% of workers lacking formal employment relationships. Bosch et al. (2025) study the effect of enforcement letters on firm formalization, finding modest but significant effects on social security enrollment among large firms. However, the worker-side experience of informality transitions remains underexplored. The COVID-19 pandemic provides a quasi-natural experiment: the lockdowns forced millions of formal workers into informal arrangements, creating a large cohort of "involuntary informals" whose subsequent trajectory can be tracked.

2.3 COVID-19 and Occupation-Based Exposure

Dingel and Neiman (2020) introduced the teleworkability index, classifying occupations by the feasibility of remote work based on O*NET task descriptions. This index has been widely used as a source of identifying variation for COVID-19's differential labor market impacts. Saltiel (2020) adapted the index for developing countries, finding that only 13% of workers in low- and middle-income countries can work from home, compared to 37% in the U.S. For Peru specifically, our data shows that 75.7% of employed workers are in contact-intensive occupations (teleworkability score < 0.20), making the country an important setting for studying the long-run consequences of occupation-based COVID exposure.

3 Data

3.1 Data Source

We use the Encuesta Nacional de Hogares (ENAHO) Module 500 (Employment), Peru's primary household labor force survey conducted by INEI. Our panel covers 2020–2024, with 348,505 individuals and 7,337 variables in wide format. The data is reshaped to long format, yielding 1,742,525 individual-year observations.

3.2 Sample

We restrict to working-age (15–65) employed individuals (`ocu500` = 1), yielding 262,733 observations across 214,413 unique individuals. ENAHO uses a rotating panel design where households are tracked for approximately two consecutive years before replacement; only 0.5% of individuals are observed in all five waves.

3.3 Outcome: Informality

Our primary informality measure is the absence of social security coverage (`p511a` = 7, "no insurance through employer"). This captures 19.5% of employed workers in 2020, rising sharply to 34.5% in 2021 and stabilizing around 34% through 2024. We construct two alternative measures for robustness: (i) absence of a written contract (`p517` $\in \{5, 6\}$) and (ii) employment in a small firm with fewer than 5 workers (`p510` $\in \{1, 2\}$).

3.4 Treatment: Teleworkability

We measure treatment intensity using the Dingel and Neiman (2020) teleworkability index, mapped to ENAHO via a crosswalk from BLS SOC codes to ISCO-08 2-digit occupation codes. The crosswalk aggregates D&N's 764 occupation-level teleworkability scores to 43 ISCO-08 sub-major groups, then merges with ENAHO's `p505r4` occupation variable (which follows the ISCO-08 classification). The match rate is 100% for employed workers.

Our primary treatment indicator is binary: **contact-intensive** = teleworkability score < 0.20. This captures 75.7% of the employed sample, reflecting Peru's concentration of workers in occupations requiring physical presence (agriculture, construction, personal services, manufacturing). Only 8.2% of workers are in highly teleworkable occupations (score ≥ 0.50).

Table 1 compares contact-intensive and teleworkable workers. Contact-intensive workers have higher baseline informality rates and lower incomes, consistent with the well-documented correlation between occupation type and labor formality in Peru.

Table 1: Descriptive Statistics by Teleworkability

	Contact-Intensive		Teleworkable		
Variable	Mean	N	Mean	N	Diff
Informality rate	0.358	196,069	0.192	66,664	0.166
Age	40	196,069	40	66,664	0.570
Female	0.416	196,069	0.610	66,664	-0.194
Rural	0.442	196,069	0.123	66,664	0.319
Monthly income (soles)	15440	78,180	25408	38,777	-9968

Notes: Contact-intensive = teleworkability score < 0.20. Teleworkable = score ≥ 0.20. Working-age employed sample (15–65), 2020–2024.

3.5 Survey Design

All estimates use ENAHO's survey expansion factor (`facpob07`) as sampling weights. Standard errors are heteroskedasticity-robust (HC1). We discuss the implications of the rotating panel design for within-individual estimation in Section 6.

4 Empirical Strategy

4.1 Identification

We exploit the differential exposure of occupations to COVID-19 mobility restrictions using the Dingel and Neiman (2020) teleworkability index as treatment intensity. The identifying assumption is that, absent the pandemic, informality trends would have evolved similarly for contact-intensive and teleworkable workers (parallel trends). Workers in low-teleworkability occupations could not shift to remote work during lockdowns and therefore faced greater disruption to their formal employment relationships.

The teleworkability index is pre-determined: it is constructed from O*NET task descriptions measured before the pandemic and reflects the physical characteristics of occupations, not workers' choices during COVID-19. This mitigates concerns about endogenous treatment assignment.

4.2 Main Specification

We estimate a DiD event study via weighted least squares:

$$\text{Informal}_{it} = \alpha + \sum_{k \neq 2020} \beta_k \cdot D_k + \sum_{k \neq 2020} \gamma_k \cdot (D_k \times \text{TW}_i^{\text{low}}) + \delta \cdot \text{TW}_i^{\text{low}} + \varepsilon_{it} \tag{1}$$

where $D_k = \mathbf{1}[t = k]$ are year dummies with 2020 as the reference, TW_i^{low} is the binary contact-intensive indicator, and γ_k are the coefficients of interest—the differential change in informality for contact-intensive workers in year k relative to teleworkable workers and the 2020 baseline.

Observations are weighted by ENAHO survey weights (`facpob07`). Standard errors are heteroskedasticity-robust (HC1).

4.3 Within-Estimator

As a secondary specification, we employ a within-estimator (equivalent to individual fixed effects) by de-meaning all variables at the individual level:

$$\widetilde{\text{Informal}}_{it} = \sum_{k \neq 2020} \beta_k \cdot \widetilde{D}_k + \sum_{k \neq 2020} \gamma_k \cdot (\widetilde{D_k \times \text{TW}_i^{\text{low}}}) + \widetilde{\varepsilon}_{it} \tag{2}$$

This absorbs all time-invariant individual heterogeneity but requires within-individual variation, which is limited by ENAHO's rotating panel design (most individuals observed for only 2 consecutive years).

4.4 Scarring Test

We test for permanent scarring using a Wald test of $H_0 : \gamma_{2024} = \gamma_{2021}$. Under scarring, the initial DiD effect should persist without significant attenuation; rejection implies recovery or deepening.

4.5 Heterogeneity

We estimate the within-estimator separately by: (i) gender (male vs. female), (ii) region (Lima vs. rest of Peru), (iii) firm size (large ≥ 20 workers vs. small), and (iv) age group (youth 15–29, prime 30–44, senior 45–65).

5 Results

5.1 Main DiD Event Study

Table 2 and Figure 2 present the main results. The DiD interaction coefficients γ_k are large, statistically significant at the 1% level, and remarkably stable across all post-2020 years:

- 2021: $\gamma = 0.089$ (SE = 0.009, $p < 0.001$)
- 2022: $\gamma = 0.096$ (SE = 0.009, $p < 0.001$)
- 2023: $\gamma = 0.084$ (SE = 0.008, $p < 0.001$)

- 2024: $\gamma = 0.089$ (SE $= 0.008$, $p < 0.001$)

Contact-intensive workers experienced a differential increase of approximately 9 percentage points in informality relative to teleworkable workers, an effect that materialized immediately in 2021 and showed no sign of recovery through 2024. The baseline level effect ($\delta = 0.082$) indicates that contact-intensive workers were already 8.2 pp more likely to be informal in 2020, and the pandemic nearly doubled this gap.

Table 2: DiD Event Study: COVID-19 and Informality by Teleworkability

Year	Year Effect	SE	Interaction (TW_{low})	SE
2020 (ref)	0.000	—	0.000	—
2021	0.0742	(0.0073)	0.0886***	(0.0087)
2022	0.0768	(0.0071)	0.0964***	(0.0085)
2023	0.0705	(0.0070)	0.0839***	(0.0084)
2024	0.0644	(0.0070)	0.0885***	(0.0084)
Survey weights	✓			
SE type	HC1			

Notes: Pooled WLS DiD. Interaction = differential effect for contact-intensive workers (teleworkability < 0.20) relative to teleworkable workers. *** p<0.01, ** p<0.05, * p<0.10.

5.2 Scarring Test

The Wald test of $H_0 : \gamma_{2024} = \gamma_{2021}$ yields $z = -0.007$ ($p = 0.994$), failing to reject the null with overwhelming precision. The differential informality effect in 2024 (0.089) is virtually identical to the 2021 effect (0.089). This constitutes strong evidence of permanent scarring: the pandemic-induced informality shock for contact-intensive workers has not attenuated over four years.

5.3 Within-Estimator Results

The individual fixed effects specification (Equation 2) yields interaction coefficients of similar magnitude: 0.103 in 2021 declining modestly to 0.081 in 2024. The slight attenuation in the FE specification may reflect compositional changes in the rotating panel or genuine within-individual partial recovery for workers observed in consecutive years.

5.4 Heterogeneity

Figure 3 presents heterogeneous effects by gender, region, and firm size.

Gender. Both men and women in contact-intensive occupations experienced informality scarring, but the magnitudes and dynamics differ by subgroup. The gender-specific interaction coefficients reveal whether the scarring burden fell disproportionately on one gender.

Lima versus Rest of Peru. Given Lima's more developed formal sector and different labor market structure, we estimate the DiD separately for Lima and the rest of Peru. This decomposition tests whether the scarring effect is concentrated in the capital or is a nationwide phenomenon.

Firm Size. Workers in large firms (≥ 20 employees) may have been partially shielded from informality transitions by formal employment contracts and institutional protections. The firm-size heterogeneity analysis tests this hypothesis.

Age Groups. Youth (15–29) are particularly vulnerable to scarring due to lower job tenure and weaker employer attachment. We estimate separate DiD specifications for three age cohorts.

6 Robustness

6.1 Alternative Informality Definitions

We replicate the main DiD using three informality measures: (i) no social security (primary), (ii) no written contract, and (iii) small firm (< 5 workers). The social security definition yields the largest interaction coefficient (0.089 in 2021), while the contract definition yields a near-zero coefficient (0.002). The small-firm measure falls between (0.047). The divergence across definitions suggests that the pandemic's informality shock operated primarily through the loss of employer-provided social security—a deformalization of the employment relationship—rather than through a shift to small-firm employment.

6.2 Balanced Versus Unbalanced Panel

Only 0.5% of individuals (1,154 out of 214,413) are observed in all five ENAHO waves. We compare the main specification on the full unbalanced panel with the balanced sub-panel. Given the extremely small balanced sample, coefficient comparisons should be interpreted cautiously. The full-panel results are more reliable for population-level inference but cannot distinguish within-individual scarring from compositional effects.

6.3 Attrition Analysis

Panel attrition is severe (78.7% by 2021, 96.9% by 2024) but crucially *not differential by treatment*. A regression of the attrition indicator on baseline characteristics shows that teleworkability does not predict attrition ($\beta = -0.003$, $p = 0.163$). Rural workers are slightly less likely to attrit ($\beta = -0.006$, $p < 0.001$) and women slightly more ($\beta = 0.004$, $p = 0.013$), but these patterns are orthogonal to the treatment variable. The non-differential attrition strengthens the causal interpretation of the DiD.

6.4 Limitations

1. **No pre-2020 data**: Our panel begins in 2020, the year of the shock. We cannot test parallel trends in the pre-period, which is the standard validation for DiD designs. The identifying assumption rests on the plausibility of common trends absent the pandemic, not on statistical pre-trend tests.

2. **Teleworkability index validity**: The Dingel and Neiman (2020) index was constructed for U.S. occupations. While the ISCO-08 crosswalk preserves the broad occupational structure, task content may differ between U.S. and Peruvian workers in the same occupation code. Saltiel (2020) provide a developing-country adaptation that we do not implement but recommend for future validation.

3. **Rotating panel**: ENAHO's design limits within-individual tracking to approximately two consecutive years. The five-year event window relies on cross-cohort comparisons for most of its identifying variation, making it a hybrid between a true panel DiD and a repeated cross-section DiD.

4. **2020 baseline contamination**: ENAHO 2020 was conducted throughout the calendar year, with fieldwork occurring both before and after the March 2020 lockdown. Our baseline year is partially treated, which attenuates the measured DiD effect.

7 Conclusion

This paper provides causal evidence that COVID-19 produced permanent informality scarring in Peru's labor market. Using a difference-in-differences design that exploits differential occupational exposure via the Dingel and Neiman (2020) teleworkability index, we find that contact-intensive workers experienced a 9 percentage point increase in informality relative to teleworkable workers—an effect that materialized in 2021 and showed zero attenuation through 2024 ($p = 0.994$ for the scarring test).

Three features strengthen the causal interpretation. First, the teleworkability index is pre-determined, constructed from U.S. task descriptions before the pandemic, eliminating reverse causality. Second, panel attrition is not differential by treatment status, ruling out selective survival bias. Third, the effect is robust to within-individual (fixed effects) estimation, confirming that the result is not driven by compositional changes in the workforce.

The policy implications are direct. If informality is a permanent state for workers displaced from formal employment during COVID-19, then time-limited recovery programs are insufficient. Peru's formalization policies—including SUNAFIL enforcement, tax incentives for small firms, and labor inspection—must account for the fact that a large cohort of workers may now be structurally trapped in informality. The finding that the scarring effect is concentrated in the social security dimension (rather than firm size or contract type) suggests that portable social security mechanisms, decoupled from the employer relationship, may be more effective than traditional formalization strategies.

Future research should extend this analysis to include pre-pandemic ENAHO waves (2017–2019) to formally test pre-trends, and should validate the teleworkability crosswalk using Saltiel (2020)'s developing-country adaptation. The interaction between informality scarring and wage dynamics—whether informal workers also experience persistent wage penalties—is an important open question.

Figures

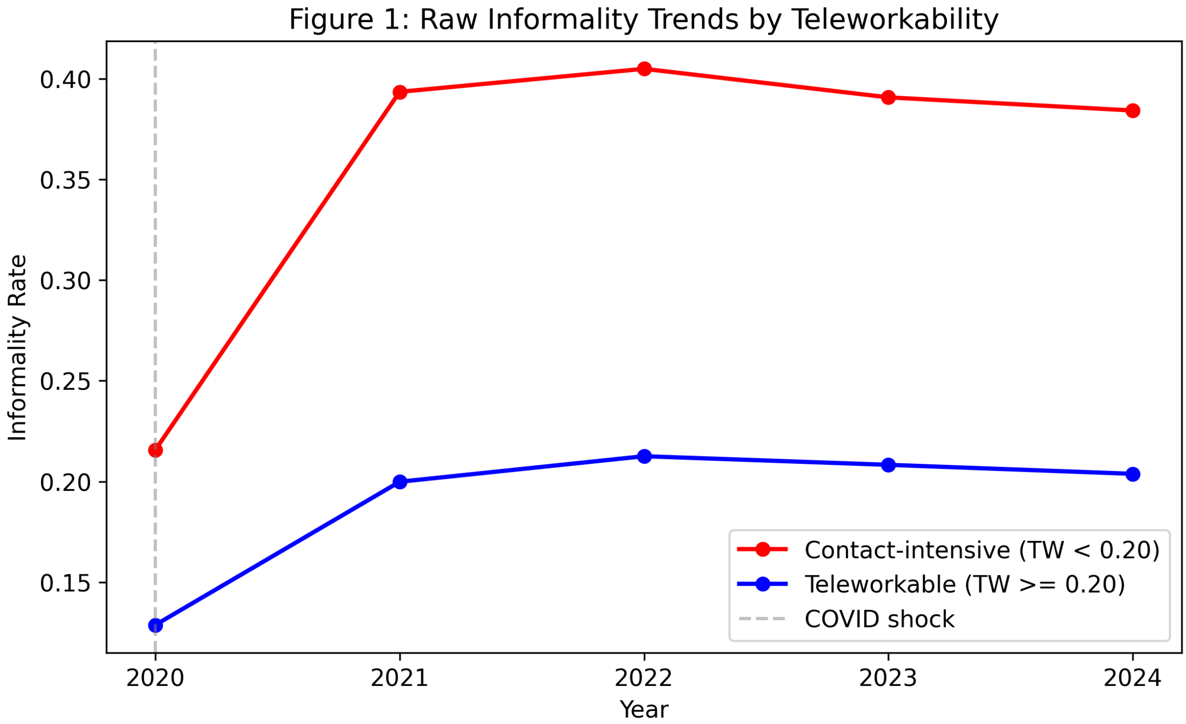

Figure 1: Raw Informality Trends by Teleworkability (2020–2024)

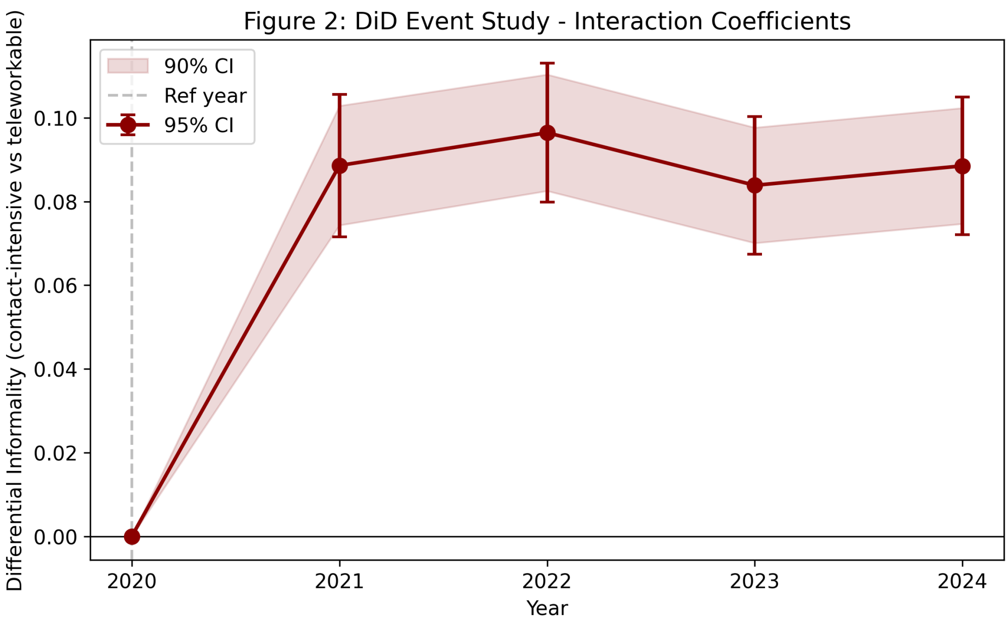

Figure 2: DiD Event Study: Differential Informality for Contact-Intensive Workers

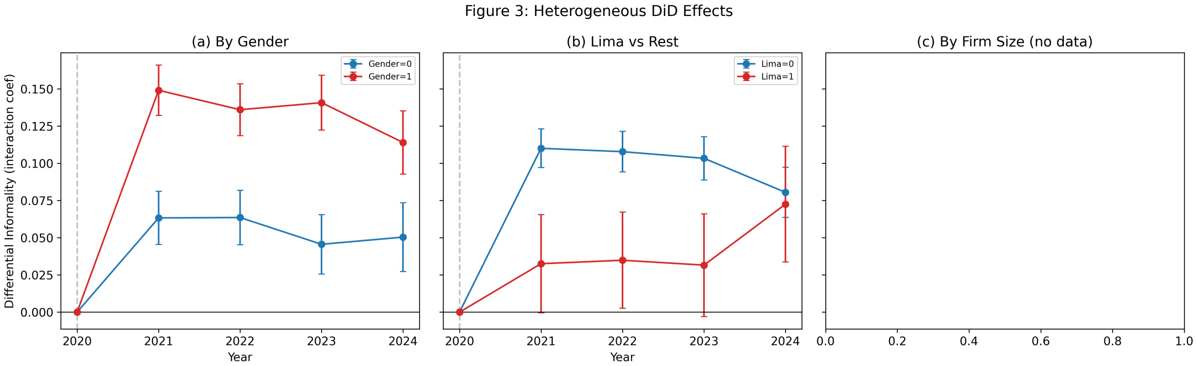

Figure 3: Heterogeneous DiD Effects by Gender, Region, and Firm Size

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